

Y-88 Source Scan: Compton-Edge Energy Calibration of the MX17

Plastic, Liquid, and SiPM-Wall Channels

n_TOF EAR2 X17 campaign, runs 224476–224479 (July 17, 2026)

Dylan Neff

analysis and document prepared with Claude (Anthropic AI assistant)

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Abstract

Four dedicated ~ 13 -minute source runs (224476–79, one per arm A/B/C/D) were taken the morning of 2026-07-17 with a single ^{88}Y source placed between the two plastic scintillator bars of one arm. ^{88}Y emits 898 and 1836 keV γ s; in organic scintillator the observable landmarks are the Compton edges at **699** and **1612 keVee**. Despite ~ 30 s of effective live time the source is bright (>3 M plastic hits on the illuminated arm), so the spectra are smooth. The runs were reprocessed privately with R. Mucciola’s PSA UserInput (WALL + LIQ pulse-shape fitting) so the liquid scintillators appear in the readout for the first time. Three results. (1) **Both plastic Compton edges are resolved on every illuminated channel**, fit as two *independent* smeared steps: the measured 1612/699 ratio (median 2.4, per channel 1.8–2.6) reproduces the expected 2.31, confirming the energy assignment from the data itself; the plastic response is linear to ~ 10 –20%. The plastic HV was clearly **raised** for these “plastic calibration” runs — the 699 edge sits at 20–27 mV, ~ 25 –39 mV/MeVee. (2) **First liquid-scintillator absolute energy scale**: the 699 keVee Compton edge is a clean bump at 22–26 mV on all four liquids (32–37 mV/MeVee), remarkably consistent arm-to-arm. (3) The same edge appears at 20–30 mV on the source-facing SiPM-wall channels (the new PSA also suppresses the wall noise hits by $\sim 10\times$). Across all three detector types the 699 keVee edge lands within 21–27 mV per arm. The exact run HV is not recorded in the data and is required to transport the plastic scale to nominal HV.

1 Runs, reprocessing, and method

The run/arm assignment is 224476=A, 224477=B, 224478=C, 224479=D (source between that arm’s plastic bars, near the middle, so both bars, the arm’s wall, and the liquid vessel behind are illuminated). The runs are beam-off (~ 1500 timer triggers, 20 ms windows), so the beam-analysis selection is inapplicable: all triggers are used and `tof/tflash` are ignored. The official processed files carry no LIQ trees (they predate the LIQ PSA UserInput), so the four runs were reprocessed with the official `RunProcessing.sh` pipeline and Mucciola’s `UserInput_2026_EAR2_X17.h` (pulse-shape fitting for WALL and LIQ); this both adds the liquids and cleans the wall reconstruction (WALL noise hits drop by $\sim 10\times$ relative to the official files). Hits are extracted with the existing cache (`fastread/extract_hits`), histogrammed in linear mV (0.2 mV bins) and converted with the per-run DAQ full-scale factors (~ 0.031 mV/ADC).

Edge extraction works on the *linear* dN/dA spectrum. The two plastic gammas produce a Compton continuum that steps down at each edge; each edge is fit *independently* as a single resolution-smeared step (complementary error function on a background), the 698.63 keVee edge as the dominant step and the weaker 1612.06 keVee edge in the window above it — so their ratio is a measured cross-check, not an input. On the SiPM walls and the liquids the 699 keVee edge instead appears as a localised bump

and is fit as a Gaussian on a linear background, the peak centre being the landmark. All errors are Poisson bootstraps (200 resamples, fixed seed) of the fine bins, floored at half a bin.

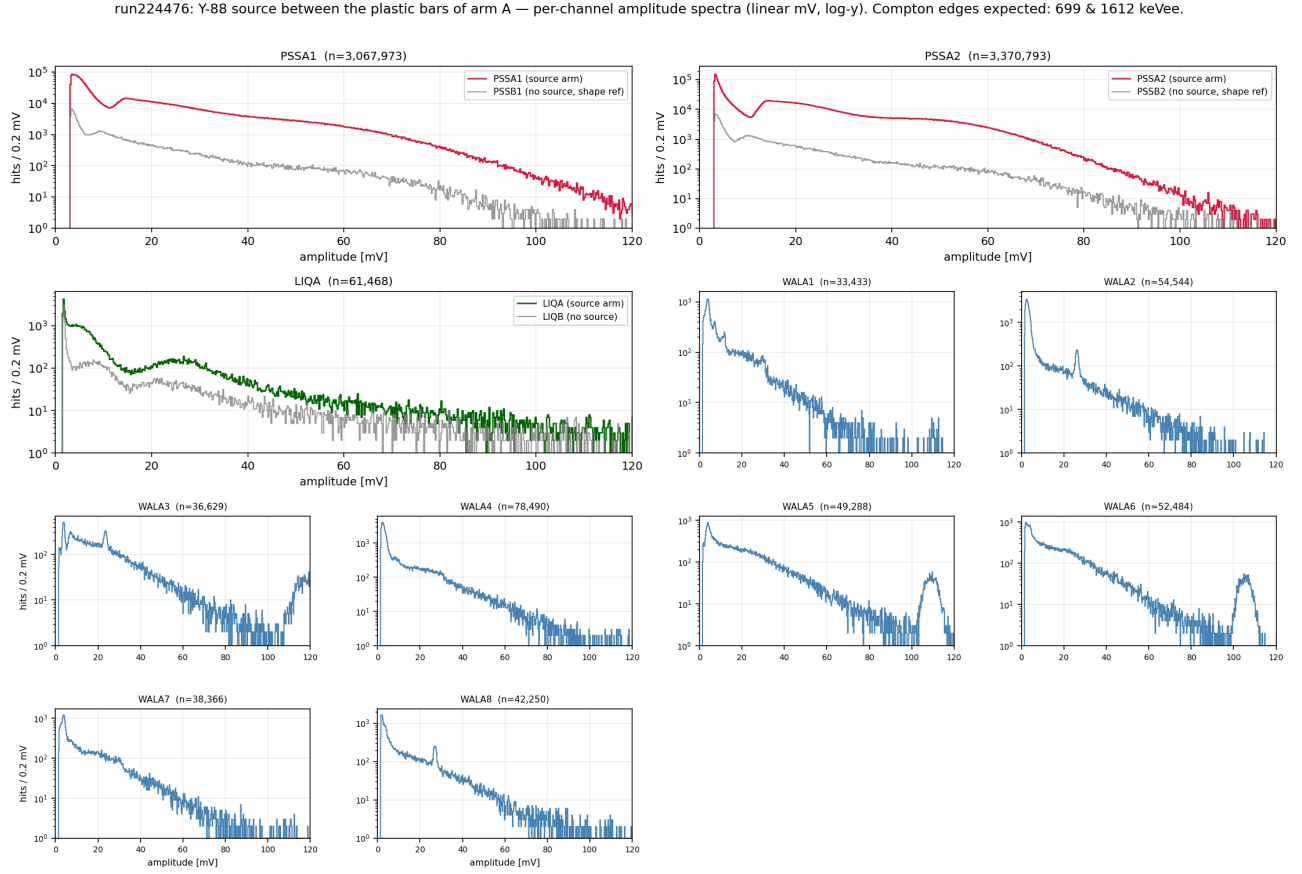


Figure 1: Run 224476 (arm A), per-channel amplitude spectra. Top: the two arm-A plastics (red) with a non-illuminated arm overlaid as a shape reference (gray). Middle-left: the arm-A liquid (green) vs. a no-source liquid — a clear Compton bump near 26 mV, the first LIQ Y-88 spectrum. Remaining panels: the eight arm-A SiPM-wall channels (now noise-suppressed by the new PSA); source-facing bars show a clean edge bump near 26 mV.

2 Plastic Compton edges and linearity

Figure 2 shows the fits; Table 1 collects the results. The 699 keVee edge (the dominant step) is the robust landmark at 20–27 mV across all eight channels; the weaker 1612 keVee edge follows at 37–64 mV. Their measured ratio (median 2.4, Table 1) reproduces the expected 2.31 without being imposed, directly confirming the two-edge interpretation; the implied nonlinearity is $\lesssim 10\text{--}20\%$. The per-channel gains span 24–39 mV/MeVee, all $\sim 20\times$ the nominal-HV MIP scale (0.65–2.05 mV/MeV), so the plastic HV was raised for these runs; the exact values are not in `DAQsettings` and are needed to transport the numbers to nominal.

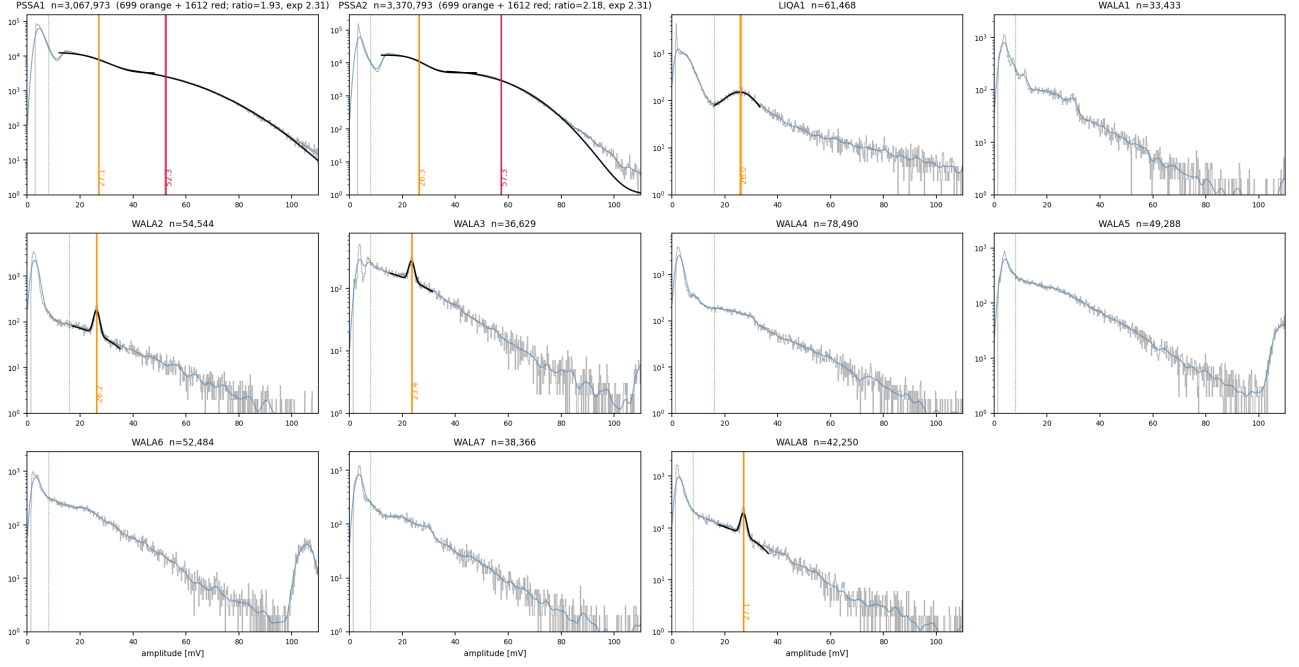


Figure 2: Arm-A edge extraction. Gray: raw linear spectrum; blue: smoothed; black: fitted model; orange/red: the 699/1612 keVee edges with bootstrap bands; green dotted: the noise valley. Plastics show the two independent step fits; the liquid and source-facing walls show the single-bump fit.

Table 1: Plastic Compton edges (mV) at the raised run HV, per channel, from all four source runs. “mV/MeVee” is the through-origin two-point slope; “ratio” is the *measured* 1612/699 (expected 2.31); “nonlin.” is ratio/2.307.

channel	699 keVee [mV]	1612 keVee [mV]	mV/MeVee	ratio	nonlin.
PSSA1	27.1	52.3	33.4	1.93	0.84
PSSA2	26.3	57.3	35.9	2.18	0.95
PSSB1	20.1	50.1	30.7	2.50	1.08
PSSB2	23.3	59.4	36.3	2.55	1.11
PSSC1	22.3	55.1	33.8	2.47	1.07
PSSC2	19.9	46.5	28.8	2.34	1.01
PSSD1	24.4	63.8	38.9	2.61	1.13
PSSD2	20.4	36.9	23.9	1.81	0.78
median	22.8	54.6	33.6	2.40	1.04

3 Liquid and SiPM-wall edges

On the liquids and walls the 699 keVee edge is a single bump (Table 2). The **liquid scale is new**: 22–26 mV (32–37 mV/MeVee) on all four vessels, consistent to $\sim 10\%$ arm-to-arm despite modest statistics (6–110 k hits). The walls give 20–30 mV on the two-to-three source-facing channels per arm; a couple of channels (e.g. WALD7) scatter high, and channels 5/6 on each arm carry an unrelated narrow peak

near 90–105 mV (an instrumental feature, not the Compton edge) that the fit ignores. Figure 3 (right) compares the 699 keVee edge across the three detector types per arm: they view the same source and, although they are different detectors at independently-set HV, all land within 21–27 mV.

The run-224404 wall-MIP cross-check from the earlier official-file pass is *not* repeated here: 224404 was reconstructed with the old PSA while these runs now use the pulse-shape-fitting PSA, so that comparison is no longer apples-to-apples. A same-PSA wall MIP (run 224489 was processed with the same UserInput) would restore it.

Table 2: Liquid and SiPM-wall 699 keVee Compton-edge bump (mV), source-facing channels. LIQ is the first absolute LIQ energy scale.

LIQ	699 [mV]	mV/MeVee	WAL	699 [mV]	mV/MeVee
LIQA	26.0	37.2	WALA2	26.2	37.5
LIQB	22.6	32.4	WALA3	23.4	33.6
LIQC	24.0	34.4	WALA8	27.1	38.7
LIQD	22.1	31.7	WALB4	26.6	38.0
			WALB5	27.0	38.6
			WALC2	21.6	30.9
			WALC3	28.7	41.1
			WALC8	24.1	34.5
			WALD2	26.5	37.9
			WALD8	19.8	28.3

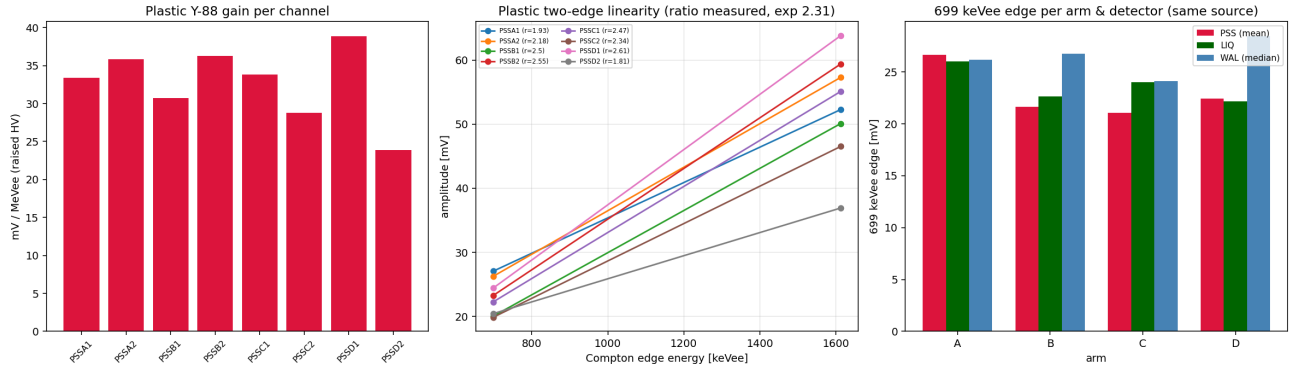


Figure 3: Left: plastic Y-88 gain per channel (mV/MeVee, raised HV). Middle: plastic two-edge linearity — the two Compton edges plus the origin, with the measured 1612/699 ratio per channel. Right: the 699 keVee edge per arm for the three detector types (same source).

4 Status and open items

The spectra, per-channel Compton edges (`calib/y88_edges*.json`) and the mV/keVee calibration (`calib/y88_energy_calib.json`) are complete for plastics, liquids, and walls on the reprocessed files. Two items remain:

1. **Plastic (and wall) HV during runs 224476–79** — not in `DAQsettings`; required to transport the plastic edges to nominal HV and compare directly with the run-224489 MIP scale.

2. **Wall MIP cross-check at the same PSA** — extract the wall MIP from run 224489 (same UserInput) to cross-check the Y-88 wall edge without the reconstruction mismatch that retired the 224404 comparison.